Algebra Monsoon 2026 Lecture 10 Stabilisers, conjugation

Let G be a group, H a subgroup.
Consider the natural action of
the group H on the set G:

$$\begin{aligned} H \times G &\longrightarrow G \\ (h, g) &\longmapsto hg. \end{aligned}$$

(restriction to H of the action of G on G by left translations).

Lemma For every $g \in G$, the H-orbit $H.g$ of g is equal to the right translate $\delta_{g*}(H)$ of H.

Proof.

$$\begin{aligned} x \in H.g &\iff x = hg \quad (\text{for some } h \in H) \\ &\iff x = \delta_g(h) \quad (\text{for some } h \in H) \\ &\iff x \in \delta_{g*}(H). \end{aligned}$$

□

Lemma $Hg_1 = Hg_2$
$\iff g_1 g_2^{-1} \in H$.

Proof If $Hg_1 = Hg_2$,
then $g_1 = hg_2$ for some $h \in H$,
so $g_1 g_2^{-1} \in H$. Conversely, if
$g_1 g_2^{-1} = h$ for some $h \in H$,
then $g_1 = hg_2$, so $g_1 \in Hg_2$,
therefore $Hg_1 = Hg_2$ as we saw
earlier. □

Summary: For every subgroup H of G,
we have the partition $H\backslash G$
by right translates of H; the
corresponding equivalence relation on G
is $g_1 \sim g_2 \iff g_1 g_2^{-1} \in H$.

Example
- If $H = \{e\}$, then $H\backslash G = G$.
- If $H = G$, then $H\backslash G = \{G\}$.

The index of a subgroup.

Let G be a group, $H \subset G$ a subgroup.

Def $(G : H) = \#\, H\backslash G$.

Example $(G : \{e\}) = \#G$, $(G : G) = 1$

Proposition $\#G = \underbrace{(G : H)}_{= \#\, H\backslash G} \#H$.

Proof As $H\backslash G$ is a partition of G, it is enough to find a bijection $H \to Hg$. Indeed, the orbital mapping $h \mapsto hg$ is surjective by definition; it is also injective because $h_1 g = h_2 g \Rightarrow h_1 = h_1$ (multiplying both sides by g^{-1}). $\square$

Corollary $|H|$ and $(G : H)$ divide $|G|$.

Corollary If $|G|$ is a prime number, then the only subgroups of G are $\{e\}$ and G.

The Stabiliser

Let G be a group, S a G-set, and $s \in S$.

Def. The **stabiliser** of s in G is

$$G_s = \{ g \in G \mid g \cdot s = s \}.$$

Proposition G_s is a subgroup of G.

Proof We have $e \in G_s$ because $e \cdot s = s$.

If $g_1, g_2 \in G_s$, so that $g_1 \cdot s = s$ and $g_2 \cdot s = s$, then

$$\begin{aligned}(g_1 g_2) \cdot s &= g_1 \cdot (g_2 \cdot s) \\ &= g_1 \cdot s \\ &= s,\end{aligned}$$

therefore $g_1 g_2 \in G_s$.

If $g \in G_s$, so that $g \cdot s = s$, then

$$\begin{aligned}s = e \cdot s &= \bar{g}^1 g \cdot s \\ &= \bar{g}^1 \cdot (g \cdot s) \\ &= \bar{g}^1 \cdot s,\end{aligned}$$

so $\bar{g}^1 \in G_s$. □

G-groups

Let G be a group.

Def. A G-group is a functor

$$BG \longrightarrow \mathcal{G}$$

into the category of groups.

As before, a G-group is a group Γ together with a morphism of groups

$$f: G \longrightarrow \mathrm{Aut}_{\mathcal{G}}\Gamma.$$

For every $g \in G$, let $\mathrm{Int}(g)$ be the mapping $h \mapsto ghg^{-1}$ from G to G.

- $\mathrm{Int}(g) \in \mathrm{Aut}_{\mathcal{G}}(G)$

Proof We have $\mathrm{Int}(g)(h_1 h_2)$

$= g(h_1 h_2)g^{-1}$

$= ghgg^{-1}h_2 g^{-1}$

$= \mathrm{Int}(g)(h_1)\,\mathrm{Int}(g)(h_2)$,

so $\mathrm{Int}(g)$ is a morphism of magmas.

Secondly $\mathrm{Int}(g) \circ \mathrm{Int}(g^{-1}) = \mathrm{Id}_G$

and $\mathrm{Int}(g^{-1}) \circ \mathrm{Int}(g) = \mathrm{Id}_G$,

so $\mathrm{Int}(g)$ is bijective.

Thus we have $\mathrm{Int}(g) \in \mathrm{Aut}_{\mathcal{G}}(G)$. □

Notation:
For $g, h \in G$, write $g_h = ghg^{-1}$ so that $g_1(g_2 h) = (g_1 g_2)h$.

It follows that

The morphism $\mathrm{Int} : G \longrightarrow \mathrm{Aut}_g(G)$.

Proposition $\mathrm{Int} : G \longrightarrow \mathrm{Aut}_g(G)$ is a morphism of groups.

Proof It is enough to show that

$$\mathrm{Int}(gh) = \mathrm{Int}(g) \circ \mathrm{Int}(h).$$

Indeed $\mathrm{Int}(gh)(x)$

$$= ghx\,h^{-1}g^{-1}$$

$$= g\,\mathrm{Int}(h)(x)\,g$$

$$= \mathrm{Int}(g)\big(\mathrm{Int}(h)(x)\big)$$

$$= \big(\mathrm{Int}(g) \circ \mathrm{Int}(h)\big)(x).$$

Hence Int is a morphism of groups. □

- The kernel of Int is $Z(G)$, the centre of the group G.

- Whenever we view G as a G-group, it will be through the morphism

$$\mathrm{Int} : G \longrightarrow \mathrm{Aut}_{\mathcal{G}}(G).$$

We say that G acts on the G-group G through conjugation.

(Earlier we considered G as a G-set under left translations).

- Every G-group can be viewed as a G-set (by composing with the forgetful functor $\mathcal{G} \longrightarrow \Xi$).

The following proposition has already been proved for arbitrary G-sets but we reprove it for the G-set G (under conjugation).

Def: Two elements $g_1, g_2 \in G$ are said to be conjugate in G if there exists $h \in G$ such that

$$g_2 = \mathrm{Int}(h)(g_1) = h g_1 h^{-1}$$

Exercise: Show that $\sigma_1 \sigma_2 \cdots \sigma_n$ is conjugate to $\sigma_r \sigma_{r+1} \cdots \sigma_n \sigma_1 \sigma_2 \cdots \sigma_{r-1}$ in the group $\langle \sigma_1, \dots, \sigma_n \rangle$.

Prop This is an equivalence relation on G.

Proof.

- (Reflexivity) We have $g = \mathrm{Int}(e)(g)$ for every $g \in G$.
- (Symmetry) If $g_2 = \mathrm{Int}(h)(g_1)$, then $g_1 = \mathrm{Int}(h^{-1})(g_2)$.
- (Transitivity) If $g_2 = \mathrm{Int}(h_1)(g_1)$ and $g_3 = \mathrm{Int}(h_2)(g_2)$,

then
$$\begin{aligned} g_3 &= \mathrm{Int}(h_2)(g_2) \\ &= h_2 g_2 h_2^{-1} \\ &= h_2 \, \mathrm{Int}(h_1)(g_1) \, h_2^{-1} \\ &= h_2 h_1 g_1 h_1^{-1} h_2^{-1} \\ &= \mathrm{Int}(h_2 h_1)(g_1). \end{aligned}$$

□

Proposition Conjugacy classes partition G.

Proof This is a special case of the fact that G-orbits partition a G-set. □

Let G be a group, $E \subset G$ a subset and $g \in G$.

Notation: $gEg^{-1} = \mathrm{Int}(g)_*(E)$.

Two subsets E_1, E_2 of G are said to be *conjugate* in G if $E_2 = gE_1g^{-1}$ for some $g \in G$.

- This is an *equivalence relation* on $\mathcal{P}(G)$.

- If E_1 is a *subgroup* of G and E_2 is *conjugate* to E_1 in G,

then $\in_2$ is a subgroup of G

(because $\mathrm{Int}(g)$ is an automorphism

of the group G for every $g \in G$.).